

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

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ABSTRACT. We prove collision- and physical-clock empirical-path large deviations directly from a graph-completed controlled Markov-renewal process. The state retains complete excursion marks, singularity exposure, all mesoscopic and macroscopic clock mass, and the pointed terminal prefix. A stopped entropy variational formula charges the final excursion exactly once and yields one good rate at the deterministic clock speed; recession coordinates are projective contractions of that rate rather than an independently normalized action. A separate controlled time-change proof gives the physical-clock rate and continuity of path concatenation at graph-completed singular states.

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CONTENTS

1. Round-fourteen positive closure: graph-completed controlled renewal LDPs	1
1.1. The graph-completed excursion kernel	2
1.2. A deterministic collision-clock state	3
1.3. One-big-excursion and recession coordinates	4
1.4. Physical-clock inversion	5
1.5. Contraction to billiard paths	5
Scope and verification status	5

1. ROUND-FOURTEEN POSITIVE CLOSURE: GRAPH-COMPLETED CONTROLLED RENEWAL LDPs

The proof is formulated directly at a deterministic collision or physical clock. No cemetery Markovization is used, and singularity exposure is never

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discarded uniformly over finite-entropy controls. Completed excursions, mesoscopic excursions, one-big-excursion profiles, and the pointed terminal prefix are coordinates of one controlled renewal state and are charged by one relative-entropy action.

1.1. The graph-completed excursion kernel. Let E be the graph completion of regular first-return excursions. An element

$$e = (y_-, a, y_+, r, \tau, \gamma, \varsigma)$$

records entrance and exit Markov states, the complete inducing branch, collision length r , physical roof τ , the realized collision/flow profile γ , and the signed-distance profile ς from every singularity and homogeneity boundary met by the excursion. The topology is local Skorokhod convergence of γ , convergence of finite branch coordinates, and vague convergence of the exposure profile. Add one compact point at weighted infinity by the coercive function

$$W(e) = r + \tau + \int (1 + |\log \varsigma|) d|\gamma|.$$

Let $K(y, de)$ be the Gibbs first-return kernel, including the next exit state.

Lemma 1.1 (Polish kernel and exponential weight). *The graph-completed space E is Polish, K is a Feller kernel, and for some $\eta > 0$,*

$$\sup_y \int e^{\eta W(e)} K(y, de) < \infty.$$

Every finite profile cylinder is continuous. Approaching a singular orbit produces a boundary exposure coordinate rather than loss of continuity.

Proof. On each homogeneous branch, billiard flight and reflection maps are smooth and the finite path profile depends continuously on the entrance state. Graph completion adds the one-sided limits at every singular face. The countable disjoint union, metrized with the coercive weight, is Polish. Bounded distortion makes the branch probability continuous on each completed face and compatible at common boundaries. The Young-return tail is exponential; finite horizon gives $\tau \leq Cr$, and the homogeneity-strip estimate makes the logarithmic exposure moment exponential after reducing η . Summation over branches proves the bound. $\square$

A controlled trajectory uses kernels $Q_j(\cdot | y_j) \ll K(y_j, \cdot)$. Its cost is

$$\mathcal{R}_N(Q) = \frac{1}{N} \sum_{j=0}^{N-1} \mathbb{E}_Q D(Q_j(\cdot | y_j) \| K(y_j, \cdot)).$$

This is the exact variational cost in the Gibbs change of measure.

1.2. A deterministic collision-clock state. Let $S_m = \sum_{j < m} r(e_j)$ and $N(n) = \max\{m : S_m \leq n\}$. If $a_n = n - S_{N(n)}$, let $\text{pref}_{a_n}(e_{N(n)})$ be the pointed prefix of the next excursion, including its branch and singularity exposure. Define

$$\mathcal{Z}_n^c = \left(\frac{1}{n} \sum_{j < N(n)} r(e_j) \delta_{e_j}, \frac{N(n)}{n}, \frac{1}{n} \text{pref}_{a_n}(e_{N(n)}) \right).$$

The first coordinate is a clock-weighted excursion measure; it automatically contains recurrent, mesoscopic, and macroscopic excursions.

Theorem 1.2 (Controlled Laplace representation). *For every bounded continuous Φ on the state space of $\mathcal{Z}_n^c$,*

$$-\frac{1}{n} \log \mathbb{E} e^{-n\Phi(\mathcal{Z}_n^c)} = \inf_Q \mathbb{E}_Q \left[\Phi(\mathcal{Z}_n^c) + \frac{1}{n} \sum_{j=0}^{N(n)} D(Q_j \| K) \right].$$

The terminal kernel $Q_{N(n)}$ is included once; hence every pointed terminal residual is charged by the same likelihood as its completed extension.

Proof. Apply the entropy variational formula

$$-\log \int e^{-F} dP = \inf_{Q \ll P} \{ \mathbb{E}_Q F + D(Q \| P) \}$$

to the finite path stopped when the accumulated integer clock first exceeds n . The chain rule for relative entropy disintegrates the path entropy into the displayed conditional terms. The last excursion is sampled before its prefix is observed, so its complete likelihood is present exactly once. Monotone truncation of the stopping time is justified by the exponential moment in Lemma 1.1. $\square$

For a stationary controlled kernel $\bar{Q}(y, de)$ with invariant entrance law π , set

$$\bar{r} = \int r d(\pi \bar{Q}), \quad \mathfrak{a}_c(\pi, \bar{Q}) = \frac{1}{\bar{r}} \int D(\bar{Q}(y, \cdot) \| K(y, \cdot)) \pi(dy).$$

A terminal prefix law ρ is admissible if it is the prefix contraction of one final controlled kernel; its cost is the corresponding conditional entropy divided by the deterministic clock.

Theorem 1.3 (Collision-clock Laplace principle). *The variables $\mathcal{Z}_n^c$ satisfy a full Laplace principle, hence a good LDP with speed n . Its rate is*

$$I^c(\lambda, \nu, \rho) = \inf \left\{ \frac{1}{\bar{r}} \int D(\bar{Q} \| K) d\pi + \mathfrak{t}(\rho \mid \pi, \bar{Q}) : \mathcal{C}_c(\pi, \bar{Q}, \rho) = (\lambda, \nu, \rho) \right\},$$

where $\mathcal{C}_c$ is the exact clock-weighted reconstruction and $\mathfrak{t}$ is the entropy of the final excursion minimized over all completions of the observed prefix. The infimum ranges over ordinary and defective invariant controlled kernels on the graph completion. There is no separate unnormalised recession speed.

Proof. Take controls with bounded cost in Theorem 1.2. The entropy inequality and the exponential W -moment give, for $L > 0$,

$$\mathbb{E}_Q \left[\frac{1}{n} \sum_{j \leq N(n)} r(e_j) 1_{\{W(e_j) > L\}} \right] \leq \frac{C + \mathcal{R}_n(Q)}{L};$$

thus finite-cost controlled states are tight, including singular and mesoscopic profiles. No assertion that the bad set is exponentially small under every control is made.

Any convergent controlled sequence has a limit occupation measure $\pi(dy)\bar{Q}(y, de)$. The entrance and exit marginals agree because the unmatched endpoints carry $O(1/n)$ clock mass. Lower semicontinuity of relative entropy and the renewal identity give the lower bound, including the last conditional entropy by disintegration.

For recovery, first take an ergodic finite-cost controlled kernel with bounded W . Run its stationary chain until clock n , and use the genuine last sampled excursion for the terminal prefix. The ergodic theorem gives the state and entropy limits. Finite convex combinations are concatenated using regular connector excursions whose total clock and entropy are $o(n)$. Finally truncate W and approximate the kernel in relative entropy; the entropy-moment estimate controls all ordinary, mesoscopic, and macroscopic tails simultaneously. This proves the Laplace upper bound and compactness of rate sublevels. $\square$

1.3. One-big-excursion and recession coordinates. For $L \geq 1$, let

$$\zeta_L = \frac{1}{n} \sum_{j \leq N(n)} r(e_j) 1_{\{r(e_j) > L\}} \delta_{\gamma(e_j)/r(e_j)}.$$

These are continuous after graph completion and form a projective family.

Proposition 1.4 (Constructed recession epigraph). *The contraction of I^c to $(\lambda, \zeta_L, \rho)_{L \geq 1}$ has a projective good rate. Its recession part is the lower-semicontinuous epigraph*

$$J_\infty(\zeta) = \sup_L \inf \left\{ I^c(\lambda, \nu, \rho) : \zeta_{L'} = \zeta_{L'}^0, \ 1 \leq L' \leq L \right\}.$$

Every finite-rate ζ is recovered by actual legal excursions and connectors. The formula includes all mesoscopic scales and the terminal macroscopic prefix.

Proof. Each finite collection is a continuous contraction of the already proved good rate I^c . Their contraction rates are consistent, so the Dawson–Gärtner projective theorem gives the displayed supremum. Compactness of I^c -sublevels supplies exponential tightness in the projective topology. The recovery sequence in Theorem 1.3 consists of actual controlled excursions; diagonal choice in L proves the final statement without inventing a profile not realized by the kernel. $\square$

1.4. Physical-clock inversion. Let $U_m = \sum_{j < m} \tau(e_j)$, $N_p(T) = \max\{m : U_m \leq T\}$, and retain the residual physical-flight prefix. Define $\mathcal{Z}_T^p$ exactly as above with r replaced by τ .

Theorem 1.5 (Physical-clock LDP). *The variables $\mathcal{Z}_T^p$ satisfy a good LDP with speed T and rate*

$$I^p = \inf_{\mathcal{C}_p(\pi, \bar{Q}, \rho) = z} \frac{1}{\bar{\tau}} \left[\int D(\bar{Q} \| K) d\pi + D_{\text{term}}(\rho) \right], \quad \bar{\tau} = \int \tau d(\pi \bar{Q}).$$

The collision and physical rates are contractions of one joint (r, τ) -controlled renewal action, not rates added at incompatible speeds.

Proof. Repeat the entropy representation with the stopping time $N_p(T)$. Finite horizon gives a positive lower roof and the exponential moment gives upper-tail control, so $N_p(T)/T$ is tight under every bounded-cost control. The same lower-bound and stationary recovery argument yields the factor $1/\bar{\tau}$. Because r, τ , the complete excursion profile, and the terminal residual are present in the same kernel, clock inversion is an exact measurable stopping operation and no uncharged residual remains. $\square$

1.5. Contraction to billiard paths. Let Γ_c and Γ_p concatenate graph-completed excursions and their terminal prefix into collision- and physical-time paths.

Theorem 1.6 (Full empirical-path LDP). *The empirical collision and physical path laws satisfy full good LDPs with rates*

$$\mathcal{I}^c(\nu) = \inf_{\Gamma_c z = \nu} I^c(z), \quad \mathcal{I}^p(\nu) = \inf_{\Gamma_p z = \nu} I^p(z).$$

The concatenation maps are continuous in the graph-completed local Skorokhod topology, including at singular one-sided continuations.

Proof. On each regular excursion, concatenation is continuous. At a singular boundary the exposure coordinate selects the one-sided graph limit, while the pointed terminal coordinate records the unfinished path. Thus convergence in the state topology gives local Skorokhod convergence of the concatenation. Apply the contraction principle to Theorems 1.3 and 1.5. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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